



Solving linear equations where brackets are used

Task A: Using reciprocals to solve equations with brackets

1) Solve these equations in two different ways. Firstly, by expanding the bracket as a first step. Secondly, by multiplying by the reciprocal.

a) $2(x + 7) = 30$

$$2(x + 7) = 30$$

b) $21 = 7(2x + 7)$

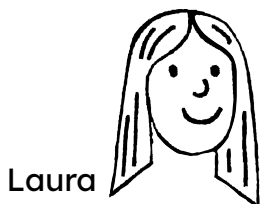
$$21 = 7(2x + 7)$$

2) Solve by expanding first:

a) $5(x - 1) = 3(3x - 11)$

b) $3(x - 1) = 5(3x - 11)$

3) Laura and Izzy have solved an equation by removing the brackets.



$$\begin{aligned} 5(x - 10) &= 45 \\ 5x - 10 &= 45 \\ 5x &= 55 \\ x &= 11 \end{aligned}$$

$$\begin{aligned} 5(x - 10) &= 45 \\ x - 10 &= 9 \\ x &= 19 \end{aligned}$$



Only one pupil is correct.

Write a sentence explaining why the right method worked and write a sentence explaining what went wrong in the other one.

Task B: Efficient methods to solve equations

1) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$5(x + 13) = 40$$

Multiplying by the reciprocal first:

$$5(x + 13) = 40$$

2) Solve both of these equations twice using two different methods.

a) $2(x + 6) = 10(x - 2)$

$$20(x + 2) = 5(7x - 1)$$

$$2(x + 6) = 10(x - 2)$$

$$20(x + 2) = 5(7x - 1)$$

3) Solve:

a) $\frac{3}{5}(7x - 4) = 21$

b) $\frac{1}{5}(5x - 15) = \frac{3}{2}(x + 4)$

